

## 37.4 Negative binomial models

The Poisson model spends its parameters on the mean and none on the variance. Whether that is enough is an empirical question, and the answer is usually no.

### 37.4.1 Where overdispersion comes from

Call a count model **overdispersed** relative to the Poisson if  $\text{Var}(Y_i) > \text{E}(Y_i)$ . Leave aside the wrong mean model, which Example (Twenty thousand records in thirty-two rows) ruled out first and which no dispersion parameter repairs. Two mechanisms remain: **unobserved heterogeneity**, the units differing in ways the regressors do not capture, which the calculation below shows always inflates the variance; and **dependence between events**, one accident making the next more likely, so that the count is not Poisson even conditionally. This section models the first.

#### Lemma 37.4.1. The negative binomial as a gamma mixture of Poissons

Let  $\Lambda$  have the gamma distribution with shape  $\kappa > 0$  and rate  $\kappa/\mu$ , so that  $\text{E}\Lambda = \mu$  and  $\text{Var}\Lambda = \mu^2/\kappa$ , and let  $Y \mid \Lambda = \lambda$  be Poisson with mean  $\lambda$ . Then:

- (a) more generally, for any mixing distribution with  $\text{E}\Lambda = \mu$  and  $\text{Var}\Lambda = \tau^2$ ,  $\text{E}Y = \mu$  and  $\text{Var}Y = \mu + \tau^2$ ;
- (b) with the gamma mixing distribution above, the marginal mass function of  $Y$  is

$$\Pr(Y = y) = \frac{\Gamma(y + \kappa)}{\Gamma(\kappa) y!} \left( \frac{\kappa}{\kappa + \mu} \right)^\kappa \left( \frac{\mu}{\kappa + \mu} \right)^y, \quad y = 0, 1, 2, \dots \quad (37.4.1)$$

with  $\text{E}Y = \mu$  and  $\text{Var}Y = \mu + \mu^2/\kappa$ ;

- (c) as  $\kappa \rightarrow \infty$  with  $\mu$  fixed, [Eq. 37.4.1](#) converges to the Poisson mass function with mean  $\mu$ .

*Proof.*

- (a)  $\text{E}Y = \text{E}\Lambda = \mu$ , and by the law of total variance, using  $\text{Var}(Y \mid \Lambda) = \Lambda$ ,  $\text{Var}Y = \text{E}\Lambda + \text{Var}\Lambda = \mu + \tau^2$ .
- (b) The gamma density is  $f(\lambda) = \{(\kappa/\mu)^\kappa / \Gamma(\kappa)\} \lambda^{\kappa-1} e^{-\kappa\lambda/\mu}$ . Therefore

$$\begin{aligned} \Pr(Y = y) &= \int_0^\infty \frac{e^{-\lambda} \lambda^y}{y!} \frac{(\kappa/\mu)^\kappa}{\Gamma(\kappa)} \lambda^{\kappa-1} e^{-\kappa\lambda/\mu} d\lambda \\ &= \frac{(\kappa/\mu)^\kappa}{y! \Gamma(\kappa)} \int_0^\infty \lambda^{y+\kappa-1} e^{-\lambda(1+\kappa/\mu)} d\lambda. \end{aligned}$$

The integral is  $\Gamma(y + \kappa)(1 + \kappa/\mu)^{-(y+\kappa)}$ , so

$$\begin{aligned} \Pr(Y = y) &= \frac{\Gamma(y + \kappa)}{\Gamma(\kappa) y!} \left( \frac{\kappa}{\mu} \right)^\kappa \left( \frac{\mu}{\mu + \kappa} \right)^{y+\kappa} \\ &= \frac{\Gamma(y + \kappa)}{\Gamma(\kappa) y!} \left( \frac{\kappa}{\kappa + \mu} \right)^\kappa \left( \frac{\mu}{\kappa + \mu} \right)^y, \end{aligned}$$

which is [Eq. 37.4.1](#). The moments follow from (a) with  $\tau^2 = \mu^2/\kappa$ .

- (c) Write  $\alpha = 1/\kappa$  and use  $\Gamma(y + \kappa)/\Gamma(\kappa) = \prod_{k=0}^{y-1} (\kappa + k) = \kappa^y \prod_{k=0}^{y-1} (1 + \alpha k)$ . Then [Eq. 37.4.1](#) becomes

$$\Pr(Y = y) = \frac{\mu^y}{y!} \left\{ \prod_{k=0}^{y-1} (1 + \alpha k) \right\} (1 + \alpha \mu)^{-1/\alpha - y}. \quad (37.4.2)$$

As  $\alpha \downarrow 0$  the product tends to one and  $(1 + \alpha \mu)^{-1/\alpha} \rightarrow e^{-\mu}$ , leaving  $e^{-\mu} \mu^y / y!$ . ■

Part (a) says the inflation is unavoidable: *any* unmodelled variation in the mean adds to the variance, whatever its shape. The gamma is chosen in (b) for its closed form, not because heterogeneity is really gamma distributed; the log-normal gives no closed form, and is the natural choice once random effects enter the picture ([Chapter 40](#)).

### Definition 37.4.1. Negative binomial regression

Let  $Y_1, \dots, Y_n$  be independent with the mass function [Eq. 37.4.1](#), means  $\mu_i$  and shapes  $\kappa_i$ , and let  $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ .

- (a) The **NB2** model takes  $\kappa_i = \kappa$  constant, so that

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\kappa}. \quad (37.4.3)$$

The variance is quadratic in the mean, and the coefficient of variation of  $\Lambda_i$  is the constant  $\kappa^{-1/2}$ .

- (b) The **NB1** model takes  $\kappa_i = \alpha \mu_i$  proportional to the mean, so that

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i}{\alpha} = \phi \mu_i, \quad \phi = 1 + \alpha^{-1}. \quad (37.4.4)$$

The variance is linear in the mean, with a constant inflation factor.

The parameter  $\alpha = 1/\kappa$  in the NB2 model is often called the **dispersion parameter**;  $\alpha = 0$  is the Poisson.

These are genuinely different models, not reparameterizations of one: NB2 says units differ multiplicatively by a fixed relative amount, NB1 that the excess variance is a fixed multiple of the mean.

## 37.4.2 Estimation and inference

### Proposition 37.4.2. Fitting the negative binomial

In the NB2 model of [Definition 37.4.1](#) with  $\kappa$  treated as a parameter:

- (a) for fixed  $\kappa$  the distribution [Eq. 37.4.1](#) is an exponential dispersion family ([Definition 34.1.1](#)) with natural parameter  $\theta = \log\{\mu/(\mu + \kappa)\}$ , cumulant function  $b(\theta) = -\kappa \log(1 - e^\theta)$ , dispersion  $\phi = 1$  and variance function  $V(\mu) = \mu + \mu^2/\kappa$ ;  
 (b) with the log link, the score for  $\boldsymbol{\beta}$  is

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \frac{y_i - \mu_i}{1 + \mu_i/\kappa} \mathbf{x}_{(i)}, \quad (37.4.5)$$

so that Fisher scoring is iteratively reweighted least squares with weights  $w_i = \mu_i/(1 + \mu_i/\kappa)$ , and the expected information for  $\boldsymbol{\beta}$  is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ ;

- (c)  $\boldsymbol{\beta}$  and  $\kappa$  are orthogonal:  $E\{\partial^2 \ell / \partial \boldsymbol{\beta} \partial \kappa\} = \mathbf{0}$ , so the expected information is block diagonal and the two may be updated alternately. Under the usual regularity conditions for maximum likelihood this also makes the asymptotic distribution of  $\hat{\boldsymbol{\beta}}$  the same whether  $\kappa$  is known or estimated (Cox and Reid 1987; not proved here);
- (d) the score for  $\kappa$  is  $\sum_i \{\psi(y_i + \kappa) - \psi(\kappa) + \log \frac{\kappa}{\kappa + \mu_i} + 1 - \frac{\kappa + y_i}{\kappa + \mu_i}\}$ , where  $\psi$  is the digamma function; it has no closed-form root and is solved numerically;
- (e) the Poisson model is the boundary case  $\alpha = 1/\kappa = 0$ . The likelihood ratio statistic  $L$  for  $H_0 : \alpha = 0$  is therefore *not* asymptotically  $\chi^2(1)$ ; under the conditions of Self and Liang (1987) its limiting distribution is  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ , so the  $p$ -value is  $\Pr\{N(0, 1) > \sqrt{L}\}$ , half the naive one;
- (f) in the NB1 model of Definition 37.4.1(b), with  $\kappa_i = \alpha\mu_i$  and a log link, the score for  $\boldsymbol{\beta}$  at fixed  $\alpha$  is

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \alpha \sum_{i=1}^n \mu_i \left\{ \psi(y_i + \alpha\mu_i) - \psi(\alpha\mu_i) + \log \frac{\alpha}{1 + \alpha} \right\} \mathbf{x}_{(i)}. \quad (37.4.6)$$

The residual  $y_i - \mu_i$  does not appear, so this is not a weighted least squares equation at all. The quasi-Poisson model of Chapter 38 shares the NB1 variance function  $V(\mu) = \phi\mu$  but has the Poisson estimating equation  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , so the two give different  $\hat{\boldsymbol{\beta}}$  from the same data (Exercise (B3)): a variance function does not determine a fit. Overdispersion is the rule rather than the exception, and modelling it changes the coefficients little and the standard errors a great deal; which of the three devices to use turns on whether a likelihood is needed, as it is for AIC and for Section 37.5.

*Proof.*

- (a) Take logarithms in Eq. 37.4.1:

$$\log \Pr(Y = y) = y \log \frac{\mu}{\mu + \kappa} + \kappa \log \frac{\kappa}{\kappa + \mu} + \log \frac{\Gamma(y + \kappa)}{\Gamma(\kappa)y!}.$$

With  $\theta = \log\{\mu/(\mu + \kappa)\}$ , which is a strictly increasing bijection from  $(0, \infty)$  onto  $(-\infty, 0)$ , we have  $\mu = \kappa e^\theta/(1 - e^\theta)$  and  $\kappa/(\kappa + \mu) = 1 - e^\theta$ , so the second term is  $\kappa \log(1 - e^\theta) = -b(\theta)$ . The last term does not involve  $\mu$ . Then  $b'(\theta) = \kappa e^\theta/(1 - e^\theta) = \mu$  and  $b''(\theta) = \kappa e^\theta/(1 - e^\theta)^2 = \mu(1 + \mu/\kappa)$ , which by Proposition 34.1.2 are the mean and the variance.

- (b) By Theorem 34.3.1 the score is  $\sum_i \mathbf{x}_{(i)}(y_i - \mu_i)(\partial \mu_i / \partial \eta_i) / V(\mu_i)$ . With the log link  $\partial \mu_i / \partial \eta_i = \mu_i$ , and  $\mu_i / V(\mu_i) = \mu_i / \{\mu_i(1 + \mu_i/\kappa)\} = 1/(1 + \mu_i/\kappa)$ , giving the score Eq. 37.4.5. The general IRLS weight of Theorem 34.4.1 is  $(\partial \mu_i / \partial \eta_i)^2 / V(\mu_i) = \mu_i/(1 + \mu_i/\kappa)$ .
- (c) Differentiating Eq. 37.4.5, written as  $\sum_i \mathbf{x}_{(i)} \kappa(y_i - \mu_i)/(\mu_i + \kappa)$ , with respect to  $\kappa$  at

fixed  $\boldsymbol{\beta}$  (so that  $\mu_i$  is fixed) gives

$$\begin{aligned}\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \kappa} &= \sum_i \mathbf{x}_{(i)}(y_i - \mu_i) \frac{\partial}{\partial \kappa} \frac{\kappa}{\mu_i + \kappa} \\ &= \sum_i \mathbf{x}_{(i)}(y_i - \mu_i) \frac{\mu_i}{(\mu_i + \kappa)^2}.\end{aligned}$$

Every term is a constant times  $y_i - \mu_i$ , which has mean zero, so the expectation vanishes. The information matrix is therefore block diagonal, and the leading block of its inverse is the inverse of the  $\boldsymbol{\beta}$  block.

- (d) Differentiate the logarithm of Eq. 37.4.1 in  $\kappa$  at fixed  $\mu$ , using  $d \log \Gamma(x)/dx = \psi(x)$ , and collect terms.
- (e) With  $\kappa_i = \alpha \mu_i$  the two ratios in Eq. 37.4.1 become constants,  $\kappa_i/(\kappa_i + \mu_i) = \alpha/(1 + \alpha)$  and  $\mu_i/(\mu_i + \kappa_i) = 1/(1 + \alpha)$ , so

$$\ell = \sum_i \left\{ \log \frac{\Gamma(y_i + \alpha \mu_i)}{\Gamma(\alpha \mu_i) y_i!} + \alpha \mu_i \log \frac{\alpha}{1 + \alpha} - y_i \log(1 + \alpha) \right\}.$$

Only  $\alpha \mu_i$  depends on  $\boldsymbol{\beta}$ ; differentiating with  $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_{(i)}$  gives Eq. 37.4.6.

- (f) By Lemma 37.4.1(c) the Poisson is the limit  $\alpha \downarrow 0$ , and  $\alpha \geq 0$  is required for Eq. 37.4.1 to be a distribution, so the null value is on the boundary. The expansion behind the  $\chi^2(1)$  limit assumes an interior maximum; here the unconstrained maximizer of the profile likelihood falls below zero about half the time, and  $L = 0$  on that event. Theorem 32.5.2 computes the analogous mixture exactly in a balanced variance-components problem; the general statement is Self and Liang's. The upper tail of  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$  at  $L$  is  $\frac{1}{2} \Pr\{\chi^2(1) > L\} = \Pr\{N(0, 1) > \sqrt{L}\}$ .

■

Part (c) makes the alternating algorithm honest as well as convenient: iterate the weighted least squares step of (b) at the current  $\kappa$ , then solve (d) at the current  $\boldsymbol{\beta}$ , and the standard errors computed as though  $\kappa$  were known are asymptotically correct.

### 37.4.3 A score test for overdispersion

The likelihood ratio test in Proposition 37.4.2(e) needs the negative binomial fit. A score test needs only the Poisson fit, which makes it the natural diagnostic to compute alongside any Poisson regression.

#### Theorem 37.4.3. Score test against NB2 alternatives

Let  $\hat{\mu}_i$  be the fitted means of a Poisson log-linear model with  $\text{rank}(\mathbf{X}) = p$ , and consider the NB2 family Eq. 37.4.2 indexed by  $\alpha \geq 0$ . Then the score for  $\alpha$  at  $\alpha = 0$  is

$$\left. \frac{\partial \ell}{\partial \alpha} \right|_{\alpha=0} = \frac{1}{2} \sum_{i=1}^n \{(y_i - \mu_i)^2 - y_i\}, \quad (37.4.7)$$

its variance under the Poisson model is  $\frac{1}{2} \sum_i \mu_i^2$ , and its covariance with the score for  $\boldsymbol{\beta}$  is zero. Suppose in addition that, as  $n \rightarrow \infty$  with  $p$  fixed, the true means stay in a fixed

compact interval inside  $(0, \infty)$  and no observation dominates, in the sense that

$$\frac{\sum_{i=1}^n \mu_i^4}{(\sum_{i=1}^n \mu_i^2)^2} \rightarrow 0. \quad (37.4.8)$$

Then

$$S = \frac{\sum_{i=1}^n \{(y_i - \hat{\mu}_i)^2 - y_i\}}{\sqrt{2 \sum_{i=1}^n \hat{\mu}_i^2}} \quad (37.4.9)$$

is asymptotically standard normal under the Poisson model, and the test rejects for large positive  $S$ .

*Proof.* Take logarithms in [Eq. 37.4.2](#):

$$\ell_i(\alpha) = y_i \log \mu_i - \log y_i! + \sum_{k=0}^{y_i-1} \log(1 + \alpha k) - \left(\frac{1}{\alpha} + y_i\right) \log(1 + \alpha \mu_i).$$

Expand in  $\alpha$ . The sum contributes  $\alpha \sum_{k=0}^{y_i-1} k + O(\alpha^2) = \alpha y_i(y_i - 1)/2 + O(\alpha^2)$ . For the last term,  $\log(1 + \alpha \mu_i) = \alpha \mu_i - \alpha^2 \mu_i^2/2 + O(\alpha^3)$ , so

$$\left(\frac{1}{\alpha} + y_i\right) \log(1 + \alpha \mu_i) = \mu_i + \alpha \left(y_i \mu_i - \frac{\mu_i^2}{2}\right) + O(\alpha^2).$$

Hence

$$\begin{aligned} \ell_i(\alpha) &= y_i \log \mu_i - \mu_i - \log y_i! \\ &\quad + \alpha \{y_i(y_i - 1)/2 - y_i \mu_i + \mu_i^2/2\} + O(\alpha^2). \end{aligned}$$

The leading term is the Poisson log-likelihood and the coefficient of  $\alpha$  is  $\frac{1}{2}\{y_i^2 - y_i - 2y_i \mu_i + \mu_i^2\} = \frac{1}{2}\{(y_i - \mu_i)^2 - y_i\}$ , which gives [Eq. 37.4.7](#).

For the variance, put  $Z_i = Y_i - \mu_i$  and  $A_i = (Y_i - \mu_i)^2 - Y_i = Z_i^2 - Z_i - \mu_i$ . For a Poisson variable,  $E Z_i = 0$ ,  $E Z_i^2 = \mu_i$ ,  $E Z_i^3 = \mu_i$  and  $E Z_i^4 = \mu_i + 3\mu_i^2$ . Therefore  $E A_i = \mu_i - 0 - \mu_i = 0$  and

$$\begin{aligned} \text{Var } A_i &= E(Z_i^2 - Z_i - \mu_i)^2 \\ &= E Z_i^4 - 2 E Z_i^3 - 2\mu_i E Z_i^2 + E Z_i^2 + 2\mu_i E Z_i + \mu_i^2 \\ &= 2\mu_i^2. \end{aligned}$$

The  $A_i$  are independent, so the variance of [Eq. 37.4.7](#) is  $\frac{1}{4} \sum_i 2\mu_i^2 = \frac{1}{2} \sum_i \mu_i^2$ .

For the covariance, differentiate [Eq. 37.4.7](#) in  $\beta$ , using  $\partial \mu_i / \partial \beta = \mu_i \mathbf{x}_{(i)}$ :

$$\frac{\partial}{\partial \beta} \frac{1}{2} \{(y_i - \mu_i)^2 - y_i\} = -\mu_i(y_i - \mu_i) \mathbf{x}_{(i)},$$

whose expectation is zero. The information between  $\alpha$  and  $\beta$  therefore vanishes at  $\alpha = 0$ , so the usual adjustment for an estimated nuisance parameter is zero and replacing  $\beta$  by  $\hat{\beta}$  does not change the asymptotic variance of the score to first order. Dividing by the standard deviation, with  $\hat{\mu}_i$  for  $\mu_i$ , gives [Eq. 37.4.9](#). For the limit, the  $A_i$  are independent with mean zero and  $E A_i^4 = O(\max(\mu_i, 1)^4)$ , a fourth-moment calculation like the one above.

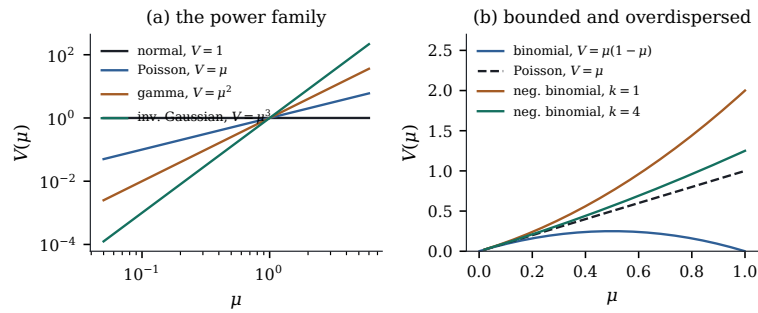


Figure 37.4.1: Sample variances of the physician-visit counts within twenty-five bins of the fitted Poisson mean, against the three fitted variance functions. The Poisson line is far below every bin.

Under the two hypotheses of the statement — means in a fixed compact interval inside  $(0, \infty)$  and [Eq. 37.4.8](#) — Lyapunov’s condition holds with  $\delta = 2$  and the standardized sum is asymptotically standard normal. Dean and Lawless (1989) treat designs in which the means are not so confined.

■

The statistic is one-sided by construction: a count *less* variable than the Poisson is possible but not negative binomial, so a large negative  $S$  points somewhere else ([Exercise \(C1\)](#)).

## 37.4.4 The physician visits again

### Example. Negative binomial fits to the visit counts

Refit the model of Example (Physician visits in a health insurance experiment) with the NB2 distribution, alternating the weighted least squares step of [Proposition 37.4.2\(b\)](#) with a one-dimensional solve for  $\kappa$ . The fit gives  $\hat{\kappa} = 0.7637$ , that is  $\hat{\alpha} = 1.3094$ ; at the overall mean of 2.860 visits the model’s variance is 13.574 against the Poisson’s 2.860. The maximized log-likelihood rises from  $-62737.33$  to  $-43468.52$ , so the likelihood ratio statistic for  $H_0 : \alpha = 0$  is 38537.6; the boundary correction of [Proposition 37.4.2\(e\)](#) halves a  $p$ -value whose logarithm is about  $-19275$ . The score statistic [Eq. 37.4.9](#), from the Poisson fit alone, is 523.0.

What changes is inference. For the individual-deductible indicator the point estimate moves only from  $-0.2648$  to  $-0.2475$ , but its standard error moves from 0.0102 to 0.0218, a factor of 2.125. Two other repairs agree: scaling the Poisson standard errors by  $\sqrt{\hat{\phi}}$ , with  $\hat{\phi} = 6.319$  the Pearson statistic over its degrees of freedom, gives 0.0257, and the sandwich estimator of Theorem 21.4.2 gives 0.0268. A Poisson analysis would have declared effects significant on the strength of a variance assumption the data reject.

Fitting NB1 instead gives log-likelihood  $-43380.59$  with  $\hat{\phi} = 4.766$ , a better fit than NB2 by AIC: 86779.2 against 86955.0, with the Poisson far behind at 125490.7. [Figure 37.4.1](#) shows why the comparison is close. Sorting the person-years by fitted mean into twenty-five bins, the bin variances follow the NB2 curve at the high end, while in the crowded low range, where most of the data and most of the likelihood sit, NB1 is closer. AIC weighs the whole likelihood and the figure the upper tail; when they disagree, choose by the use the model will be put to.

```

import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats

data = sm.datasets.randhie.load_pandas().data
y = data["mdvis"].to_numpy(float)
names = ["intercept", "lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"
        ]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names
        [1:]])
n, p = X.shape

beta_pois = np.zeros(p)
beta_pois[0] = np.log(y.mean())
for _ in range(50):
    mu = np.exp(X @ beta_pois)
    XW = X * mu[:, None]
    beta_pois = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta_pois + (y - mu) / mu))
mu_pois = np.exp(X @ beta_pois)
info_pois = X.T @ (X * mu_pois[:, None])
se_pois = np.sqrt(np.diag(np.linalg.inv(info_pois)))
log_factorial = special.gammaln(y + 1)
loglik_pois = np.sum(y * np.log(mu_pois) - mu_pois - log_factorial)
print(f"Poisson log-likelihood {loglik_pois:.2f}")

```

[Run this code online](#)

```

def nb_loglik(beta, kappa, X, y):
    """Negative binomial log-likelihood: mean exp(X beta), shape kappa (NB2 if
    scalar)."""
    mu = np.exp(X @ beta)
    return np.sum(special.gammaln(y + kappa) - special.gammaln(kappa)
                  - special.gammaln(y + 1)
                  + kappa * np.log(kappa / (kappa + mu))
                  + y * np.log(mu / (mu + kappa)))

def kappa_score(kappa, mu, y):
    """Derivative of the log-likelihood in the shape parameter."""
    return np.sum(special.digamma(y + kappa) - special.digamma(kappa)
                  + np.log(kappa / (kappa + mu)) + 1 - (kappa + y) / (kappa + mu))

def fit_nb2(X, y, tol=1e-11, maxit=200):
    """Alternate IRLS in beta with a one-dimensional search in kappa."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    kappa = 1.0
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        w = mu / (1 + mu / kappa) # NB2 working weights, log link
        XW = X * w[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
        new_kappa = optimize.brentq(kappa_score, 1e-4, 1e4,
                                   args=(np.exp(X @ step), y), xtol=1e-12)
        done = max(np.abs(step - beta).max(), abs(new_kappa - kappa)) < tol
        beta, kappa = step, new_kappa
    if done:

```

```

        break
    return beta, kappa, nb_loglik(beta, kappa, X, y)

beta_nb2, kappa, loglik_nb2 = fit_nb2(X, y)
mu_nb2 = np.exp(X @ beta_nb2)
weights = mu_nb2 / (1 + mu_nb2 / kappa)          # IRLS weights for NB2 with a log
link
se_nb2 = np.sqrt(np.diag(np.linalg.inv(X.T @ (X * weights[:, None]))))

print(f"kappa = {kappa:.4f}    log-likelihood {loglik_nb2:.2f}")
for name, b, s, sp in zip(names, beta_nb2, se_nb2, se_pois):
    print(f"{name:10s} {b:8.4f}    se {s:.4f}    (Poisson se {sp:.4f})")

```

[Run this code online](#)

```

lr = 2 * (loglik_nb2 - loglik_pois)
# the null puts 1/kappa on the boundary, so the null distribution of L is the
mixture
# (1/2) chi2(0) + (1/2) chi2(1) and the p-value is the normal tail at sqrt(L)
log_p = stats.norm.logsf(np.sqrt(lr))

score = np.sum((y - mu_pois) ** 2 - y) / np.sqrt(2 * np.sum(mu_pois ** 2))
print(f"likelihood ratio {lr:.1f}, log p-value {log_p:.0f}")
print(f"score statistic {score:.1f}, log p-value {stats.norm.logsf(score):.0f}")

```

[Run this code online](#)

## 37.4.5 Exercises

### 37.4.5.1 A. Check your understanding

#### Exercise. A1

A negative binomial NB2 fit reports  $\hat{\kappa} = 2$ . By what factor does the model inflate the variance over the Poisson at a fitted mean of 1, of 10, and of 100? Repeat for an NB1 fit with  $\hat{\phi} = 3$ , and comment.

#### Exercise. A2

A colleague reports a likelihood ratio statistic of 2.9 for Poisson against NB2 and, comparing with  $\chi^2(1)$ , obtains  $p = 0.089$  and concludes there is no overdispersion. What is the correct  $p$ -value, and does the conclusion change at the five per cent level?

### 37.4.5.2 B. Practice

#### Exercise. B1

For an intercept-only Poisson model,  $\hat{\mu}_i = \bar{y}$  for every  $i$ . Show that the score statistic Eq. 37.4.9 reduces to  $\sqrt{n/2} \{s_n^2/\bar{y} - 1\}$  with  $s_n^2 = n^{-1} \sum_i (y_i - \bar{y})^2$ , so that it compares the sample variance with the sample mean. Evaluate it for  $n = 100$ ,  $\bar{y} = 3$ ,  $s_n^2 = 5$ .

*Solution.* With  $\hat{\mu}_i = \bar{y}$  the numerator is  $\sum_i (y_i - \bar{y})^2 - n\bar{y} = n(s_n^2 - \bar{y})$  and the denominator is  $\sqrt{2n\bar{y}^2} = \bar{y}\sqrt{2n}$ , so the ratio is  $\sqrt{n/2}(s_n^2 - \bar{y})/\bar{y}$ . For the numbers given this is  $\sqrt{50} \times (5 - 3)/3$ , about 4.7: strong evidence of overdispersion.

### Exercise. B2

Show that if  $Y$  is NB2 with mean  $\mu$  and shape  $\kappa$ , then  $\Pr(Y = 0) = (1 + \mu/\kappa)^{-\kappa}$ , that this exceeds  $e^{-\mu}$  for every finite  $\kappa$ , and that it tends to one as  $\kappa \rightarrow 0$  with  $\mu$  fixed. Interpret the last statement.

*Solution.* Setting  $y = 0$  in Eq. 37.4.1 leaves  $\{\kappa/(\kappa + \mu)\}^\kappa$ , which is  $(1 + \mu/\kappa)^{-\kappa}$ . Since  $\log(1 + x) < x$  for  $x > 0$ ,  $\kappa \log(1 + \mu/\kappa) < \mu$ , so the probability exceeds  $e^{-\mu}$ . As  $\kappa \rightarrow 0$ ,  $\kappa \log(1 + \mu/\kappa) \rightarrow 0$  and the probability tends to one: extreme heterogeneity concentrates the distribution on zero with occasional very large counts, which is why the negative binomial can on its own account for a great many zeros.

### Exercise. B3

Show that the quasi-Poisson estimating equation for  $\beta$  is the *same* as the Poisson one, so that  $\hat{\beta}$  is unchanged, while the NB2 equation Eq. 37.4.5 downweights observations with large fitted means. Which model would you expect to be more affected by a single very large count?

### Exercise. B4

Show that the deviance (Definition 34.5.1) of the NB2 model with  $\kappa$  known is

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i + \kappa) \log \frac{y_i + \kappa}{\hat{\mu}_i + \kappa} \right\},$$

and that it tends to the Poisson deviance Eq. 37.1.5 as  $\kappa \rightarrow \infty$ . Why does an analysis of deviance based on it need  $\kappa$  to be held fixed across the models compared?

*Solution.* Dropping the terms free of  $\mu$ , the log-likelihood contribution of observation  $i$  is  $y_i \log\{\mu_i/(\mu_i + \kappa)\} + \kappa \log\{\kappa/(\mu_i + \kappa)\}$ . The saturated model puts  $\mu_i = y_i$ , so twice the difference is

$$2 \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} + (y_i + \kappa) \log \frac{\hat{\mu}_i + \kappa}{y_i + \kappa} \right\},$$

which is the stated formula. For the limit write  $(y_i + \kappa) \log\{1 + (y_i - \hat{\mu}_i)/(\hat{\mu}_i + \kappa)\} = (y_i - \hat{\mu}_i) + O(\kappa^{-1})$ . The deviance depends on  $\kappa$ , so two models fitted with different  $\hat{\kappa}$  have deviances on different scales and their difference is not a likelihood ratio statistic; compare their maximized log-likelihoods instead.

### 37.4.5.3 C. Going deeper

**Exercise. C1**

Construct a count distribution with mean  $\mu$  and variance strictly less than  $\mu$ , and explain why no member of the negative binomial family has this property. What mechanisms produce underdispersed counts in practice, and what would you fit?

*Solution.* The binomial with  $m$  trials and probability  $\pi$  has mean  $m\pi$  and variance  $m\pi(1 - \pi) < m\pi$ , while by [Lemma 37.4.1\(a\)](#) every Poisson mixture has variance at least its mean. Underdispersion arises when events are regular rather than random — inspections on a schedule, quotas — and the natural models are the binomial, or a Conway–Maxwell–Poisson family with a dispersion parameter free on both sides of one.

**Exercise. C2**

[Proposition 37.4.2\(a\)](#) identifies the canonical link of the NB2 family as  $\theta = \log\{\mu/(\mu + \kappa)\}$ . Work out the likelihood equations under this link, show that they take the form  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , and explain why the log link is nevertheless the standard choice.